



Terms in Probability

Experiment – is a controlled study whose outcome is uncertain but not entirely unknown.

Trial – a recorded result of an experiment.

Outcome – is one of the possible results from an experiment trial. It also refers to the basic unit of possible occurrences. For example, in tossing a coin, the outcomes are heads and tails.

Event – is some combination of possible outcomes in one experiment trial. In picking a card from an ordinary deck of cards, the outcomes are the 52 cards themselves. Examples of events are:

“Pick a 5” (one of the four cards numbered 5)

“Pick a heart” (one of the 13 cards in the suit of heart)

“Pick a face card (one of the 12 cards showing a face i.e. king, queen or jack)

Frequency of the outcome – refers to the number of times a certain outcome will occur. For example, if you roll a die 50 times, you are conducting 50 trials of the same experiment. The number of times a “6” comes up is known as the frequency of the outcome.

The **relative frequency of the outcome** is the ratio of the frequency of the outcome to the number of trials. Mathematically,

$$RF = \frac{\text{no. of occurrences}}{\text{no. of trials}}$$

Mutually Exclusive

Two or more events are said to be **mutually exclusive** if no two of them can possibly happen in the same trial.

For example, in throw a die, it is not possible to have one and six face up at the same time or in a draw of a single card from a deck of cards, you can't pick a heart and a diamond at the same time



If two or more events are mutually exclusive, the probability that either one or the other will occur is the **sum of their probabilities**.

The probability for a mutually exclusive event (E or F) is:

$$P_{E \text{ or } F} = P_E + P_F$$

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Independent Events

An event is **independent** if the outcome of one trial has no effect on the outcome of any other trial.

A good example is when a coin is tossed, the outcome has nothing to do with what happened on the previous toss and will not affect the next one.

If two events are independent, the probability that the two events occur is the **product of their individual probabilities**.

The probability of **independent events** (E and F) is:

$$P_{E \text{ \& } F} = P_E \times P_F$$

BINOMIAL DISTRIBUTION

If there are two possible outcomes of an event and the possibilities of the outcome are independent and constant, the distribution of probabilities is called **binomial distribution**.

Binomial or repeated trial probability:

$$P = {}_n C_r p^r q^{n-r}$$

where:

p = probability of success

q = probability of failure = 1 – p

n = number of trials

r = number of successful trials

What is an Odd?

The term “**odd**” is an ambiguous word that may refer to the probability that an event occurs, or it can be used to indicate the payoff on a winning bet. It sometimes may mean or is synonymous to the word “**chance**”.

If a coin is tossed, the probability of coming up head is 50% and the probability of coming up tail is also 50%. This is an example of 1-to-1 odds or even odds. For a game with even odds, a one peso bet is paid exactly one peso upon winning. This is known as the “**true odds**”. However, there are instances that 90 centavos is paid instead to a one peso bet, which would give it an advantage on each game. This refers to the “**payoff odds**”. The payoff odds are not always the same as the true odds.

The odds of an event, E occurs is the ratio of the probability that it will occur to the probability that it will not occur.

$$\begin{aligned} \text{odds for E} &= \frac{\text{prob. that E occurs}}{\text{prob. that E does not occur}} \\ \text{odds for event E} &= \frac{P_E}{1 - P_E} \end{aligned}$$

Problem:

What are the odds for throwing a total of 5 and 10 in rolling two dice?

Solution:

First, solve the probability of throwing a total of 5: For a total of 5, it must be (1 and 4), (4 and 1), (2 and 3) and (3 and 2).

$$P_5 = \frac{4}{36} = \frac{1}{9}$$

Then, solve the probability of throwing a total of 10: For a total of 10, it must be (4 and 6), (6 and 4), and (5 and 5).

$$\begin{aligned} P_{10} &= \frac{3}{36} = \frac{1}{12} \\ P_{5 \text{ or } 10} &= \frac{1}{9} + \frac{1}{12} = \frac{7}{36} \end{aligned}$$

The odds for throwing a total of 5 or 10 is:

$$\text{odd} = \frac{\frac{7}{36}}{1 - \frac{7}{36}} = \frac{7}{29}$$

This is expressed as **7-to-29 odds**.

Odd for and Odd against

If the “**odds for**” an event are given as “a” and “b”, then the probability of that event would be:

$$\text{odds for an event} = \frac{a}{a+b}$$

The “**odds against**” an event are the reciprocal of the “odds for” the event. The odds of 5 to 7 for an event is translated into odds of 7 to 5 against it.

Problem:

A player in a certain card game lays odd of 3 to 5 for him to win. What would be the payoff for a bet of 3 pesos?

Solution:

$$\text{Odds for} = \frac{3}{3+5} = \frac{3}{8}$$

Therefore, it is a 3-to-8 odd. A bet of 3 pesos will have a payoff of 8 pesos.



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